Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Develop Blockchain and Web3 App with Vibe Coding

TGS Ref No: TGS-2021009338

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version v18.1

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
17.0	Original source	Original trainer slides and approved assessment content	Tertiary Infotech Academy
v18.0	13 September 2026	Technical rebuild, 12 executable activities and visual source mapping	Tertiary Infotech Academy
v18.1	13 September 2026	Kept legacy procedures in notes/LG; concept-led slides and aligned final release	Tertiary Infotech Academy

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TABLE OF CONTENTS

Right-click and choose “Update Field”, or press F9, to build the table of contents.

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Course delivery and assessment

Develop Blockchain and Web3 App with Vibe Coding (TGS-2021009338) is delivered across two days and 16 programme hours (13.5 training and 2.5 assessment). Approved assessment plan durations are WA60 minutes and PP90 minutes separately. Public-page assessment-hours discrepancy requires provider reconciliation; the original approved instruments are retained.

Learning outcomes and mapping

LO1: Analyze business requirements for Blockchain, Cryptocurrency, Smart Contract and Token Economics.

LO2: Code smart contract for Ethereum blockchain using Solidity programming.

LO3: Test smart contract and monitor its outputs.

LO4: Deploy smart contract and ensure coding documentation is accurate.

Topic 1 Blockchain, Cryptocurrency, Smart Contract and Token Economics — A1, K1, K2

Topic 2 Ethereum Blockchain and Solidity Programming — A2, A3, K3

Topic 3 Smart Contract Testing — A4, A5, K4

Topic 4 Smart Contract Deployment — A6, A7, K5

Day 1 instructional schedule

Session	Minutes	Slides	Method and evidence
Attendance / learner briefing	30	2–13	Trainer verifies identity, download access and learning goals.
Topic 1 mechanism instruction	120	13–75	Source material, exact contracts, architecture, failure controls and observable evidence.
Topic 2 mechanism instruction	120	76–262	Source material, exact contracts, architecture, failure controls and observable evidence.
Activity 01 Hash and anchor certificate evidence	35	65–69	Individual build/test evidence, pair code review and requirement mapping.
Activity 02 Map buyer requirements to Ether escrow	35	70–74	Individual build/test evidence, pair code review and requirement mapping.
Activity 03 Compile and review a supplier purchase order	35	153–157	Individual build/test evidence, pair code review and requirement

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			mapping.
Activity 04 Issue and transfer ERC-20 workshop credits	35	158–162	Individual build/test evidence, pair code review and requirement mapping.
Activity 05 Issue and revoke ERC-721 course certificates	35	163–167	Individual build/test evidence, pair code review and requirement mapping.
Activity 06 Track custody for pharmaceutical batches	35	168–172	Individual build/test evidence, pair code review and requirement mapping.
Evidence handover and consolidation	0	262–264	Reconcile requirements, outputs, documentation and remaining verification boundaries.

Instructional total: 480 minutes. Day1:480 minutes instruction; Day2:330 minutes instruction +150 assessment. Lunch and comfort breaks are scheduled outside programme contact time. Trainer records AM/PM digital attendance.

Day 2 instructional schedule

Session	Minutes	Slides	Method and evidence
Attendance / learner briefing	30	2–13	Trainer verifies identity, download access and learning goals.
Topic 3 mechanism instruction	30	174–227	Source material, exact contracts, architecture, failure controls and observable evidence.
Topic 4 mechanism instruction	30	228–263	Source material, exact contracts, architecture, failure controls and observable evidence.
Activity 07 Test eligible voting and duplicate controls	40	207–211	Individual build/test evidence, pair code review and requirement mapping.
Activity 08 Verify withdrawal controls against reentrancy	40	212–216	Individual build/test evidence, pair code review and requirement mapping.
Activity 09 Measure gas for storage transition classes	40	217–221	Individual build/test evidence, pair code review and requirement mapping.
Activity 10 Index receipt events and reconcile records	40	222–226	Individual build/test evidence, pair code review and requirement

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			mapping.
Activity 11 Publish a reproducible local deployment manifest	40	251–255	Individual build/test evidence, pair code review and requirement mapping.
Activity 12 Connect a wallet and verify a certificate dApp	40	256–260	Individual build/test evidence, pair code review and requirement mapping.
Evidence handover and consolidation	0	262–264	Reconcile requirements, outputs, documentation and remaining verification boundaries.

Instructional total: 330 minutes. Day1:480 minutes instruction; Day2:330 minutes instruction +150 assessment. Lunch and comfort breaks are scheduled outside programme contact time. Trainer records AM/PM digital attendance.

Assessment schedule and sign off

WA: 60 minutes, five short-answer questions, K1–K5. PP: 90 minutes, four tasks: Task1 A1; Task2 A2/A3; Task3 A4/A5; Task4 A6/A7. Total150 assessment minutes within16 programme hours. Briefing precedes assessment. Closing slide sequence: Assessment → Assessment Flow → Digital Attendance/TRAQOM → Thank You.

Trainer evidence and boundaries

Local Py-EVM code execution verifies opcode-level contract behaviour. Browser wallet, public Sepolia, explorer and external dApps must be separately exercised and evidenced when required. Preserve current manifests, receipts, test logs and source locks. Do not treat compiler success, wallet approval or a transaction hash alone as successful runtime/deployment.

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